BAGUETTE: Bayesian Analysis of Gene Expression Transition Identification from Spatially Resolved Transcriptomics

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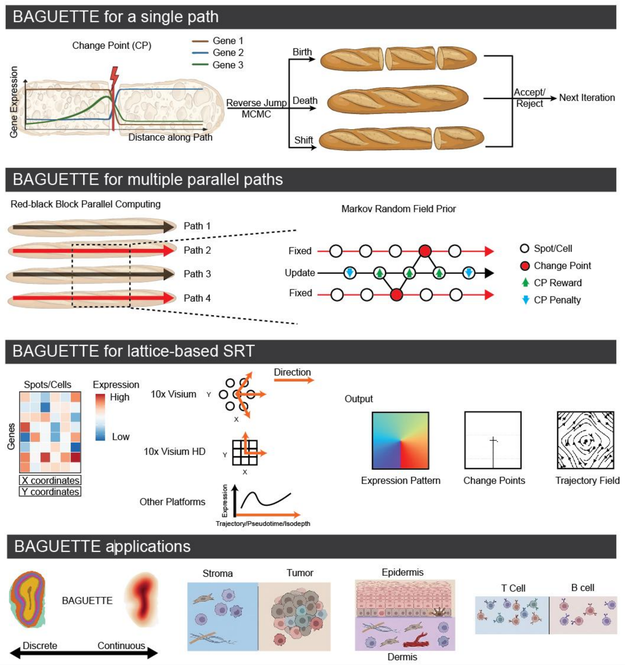
1. INTRODUCTION

- **Motivation**

  - Elevating boundary modeling as the primary goal fills the gap between fully discrete spatial domains and smooth continuous assumptions.



2. METHOD

- **Zero-inflated negative binomial model along one path**

Given a line ℓ and a gene subset $\mathcal{G} \subseteq \{1, \dots, p\}$, gene expressions are modeled via

$$y_{\ell ij} \sim \begin{cases} 0, & \eta_{\ell ij} = 1, \\ \mathrm{NB}(\mu_{\ell ij}, \phi_{\ell j}), & \eta_{\ell ij} = 0, \end{cases} \qquad \eta_{\ell ij} \sim \mathrm{Bern}(\pi_\ell)$$

- $i \in \{1, \dots, n_l\}$: spot;
- $j \in \mathcal{G}$: gene;
- $\mu_{\ell ij}$: NB mean with $E = \mu_{\ell ij}$;
- $\phi_{\ell j} > 0$: NB size with $\mathrm{Var} = \mu_{\ell ij} + \mu_{\ell ij}^2/\phi_{\ell j}$;
- $\eta_{\ell ij} \in \{0, 1\}$: indicator of structure zero;
- $\pi_\ell \in [0, 1]$: probability of structure zero.

Piecewise linear modeling of the expression mean:

$$\log \mu_{\ell ij} = \log s_{\ell i} + \beta^{(0)}_{\ell \kappa_{\ell ij}} + \beta^{(1)}_{\ell \kappa_{\ell ij}} t_{\ell i},$$

- $s_{\ell i}$: spot-level nuisance offsets, e.g., library size factor $s_{\ell i} \propto \sum_{j=1}^{p} y_{\ell ij}$;
- $\boldsymbol{\beta}_{\ell \kappa_{\ell ij}} = \left\{\beta^{(0)}_{\ell \kappa_{\ell ij}}, \beta^{(1)}_{\ell \kappa_{\ell ij}}\right\}$: intercept and slope for each point;
- $t_{\ell i}$: 1 dimensional coordinates along the line.

Change points:

For line ℓ,

- Define the **change-point indices** $\boldsymbol{\tau}_\ell = \{\tau_{\ell,1}, \dots, \tau_{\ell,K_\ell}\} \in \mathcal{S}_\ell$, where $\mathcal{S}_\ell$ is the set of all valid configurations satisfying boundary and spacing constraints.
- The corresponding **indicator vector** $\boldsymbol{\delta}_\ell = (\delta_{\ell 1}, \dots, \delta_{\ell n_\ell})^\top$ shows $\delta_{\ell i} = \mathbb{I}(i \in \boldsymbol{\tau}_\ell)$.
- Inducing **segmentation vector** $\boldsymbol{\kappa}_\ell = (\kappa_{\ell 1}, \dots, \kappa_{\ell n_\ell})^\top$, where $\kappa_{\ell i} = \sum_{t=1}^{i} \delta_{\ell t}$.

ℓ	i=1	2	3	4	5	6	7	8	9
$\boldsymbol{\tau}_\ell$				4			7		
$\boldsymbol{\delta}_\ell$	0	0	0	1	0	0	1	0	0
$\boldsymbol{\kappa}_\ell$	0	0	0	1	1	1	2	2	2

Prior:

$$K_\ell \sim \mathrm{TruncPoi}(\lambda, K_{\max}) \qquad p(\boldsymbol{\tau}_\ell \mid K_\ell) \propto \mathbb{I}(\boldsymbol{\tau}_\ell \in \mathcal{S}_\ell)$$

$$\boldsymbol{\beta}_{\ell kj} \sim \mathcal{N}(\mathbf{0}, \sigma_\beta^2 \mathbf{I}) \qquad \phi_{\ell j} \sim \mathrm{Gamma}(a_\phi, b_\phi) \qquad \pi_\ell \sim \mathrm{Beta}(a_\pi, b_\pi)$$

- **Block Gibbs sampler**

$$\boldsymbol{\Theta} = \left\{ \underbrace{\boldsymbol{\tau}_\ell}_{\text{Change Points}}, \underbrace{\{\boldsymbol{\beta}_{\ell kj}\}_{k=1,\dots,K_\ell+1; j\in\mathcal{G}}}_{\text{Segment Regression Coeffs}}, \underbrace{\{\phi_{\ell j}\}_{j\in\mathcal{G}}}_{\text{Size}}, \underbrace{\pi_\ell}_{\text{Zero Prob.}}, \underbrace{\{\eta_{\ell ij}\}_{i=1,\dots,n_\ell; j\in\mathcal{G}}}_{\text{Latent Zero Indicators}} \right\}$$

- **Block 1: updating NB part**

$$p(\boldsymbol{\tau}_\ell, \boldsymbol{\beta} \mid \mathbf{Y}_\ell, \boldsymbol{\eta}_\ell, \boldsymbol{\phi}_\ell) \propto \quad p(\mathbf{Y}_\ell \mid \boldsymbol{\tau}_\ell, \boldsymbol{\beta}, \boldsymbol{\phi}_\ell, \boldsymbol{\eta}_\ell) \cdot p(\boldsymbol{\tau}_\ell \mid K_\ell)\, p(K_\ell) \cdot p(\boldsymbol{\beta})$$

$$p(\boldsymbol{\phi}_\ell \mid \mathbf{Y}_\ell, \boldsymbol{\eta}_\ell, \boldsymbol{\tau}_\ell, \boldsymbol{\beta}) \propto \quad p(\mathbf{Y}_\ell \mid \boldsymbol{\tau}_\ell, \boldsymbol{\beta}, \boldsymbol{\phi}_\ell, \boldsymbol{\eta}_\ell) \cdot p(\boldsymbol{\phi}_\ell)$$

- **Block 2: updating Zero Inflation part**

$$P(\eta_{\ell ij} = 1 \mid y_{\ell ij} = 0, \mu_{\ell ij}, \phi_{\ell j}, \pi_\ell) = \frac{\pi_\ell}{\pi_\ell + (1 - \pi_\ell) \cdot \mathrm{NB}(0 \mid \mu_{\ell ij}, \phi_{\ell j})}.$$

$$\pi_\ell \mid \boldsymbol{\eta}_\ell \sim \mathrm{Beta}\left(a_\pi + \sum_{i,j} \eta_{\ell ij}, \quad b_\pi + n_\ell \times |\mathcal{G}| - \sum_{i,j} \eta_{\ell ij}\right).$$

- **Posterior**

1. MCMC Samples: Let $\mathcal{T}_\ell = \{\boldsymbol{\tau}_\ell^{(1)}, \dots, \boldsymbol{\tau}_\ell^{(B)}\}$ be post burn-in samples.

2. Point Estimate: Let $\mathcal{T}_\ell^U = \mathrm{unique}(\mathcal{T}_\ell) = \{\boldsymbol{\tau}_\ell^1, \dots, \boldsymbol{\tau}_\ell^U\} \subseteq \mathcal{S}_\ell$ with empirical posterior weights $\hat{w}_u \propto \#\{b : \boldsymbol{\tau}_\ell^{(b)} = \boldsymbol{\tau}_\ell^u\}$. The medoid is

$$\hat{\boldsymbol{\tau}}_{\mathrm{med}} = \arg\min_{\boldsymbol{\tau} \in \mathcal{S}_\ell} \mathbb{E}_{\boldsymbol{\tau}_\ell \mid \mathbf{Y}_\ell}\left[d(\boldsymbol{\tau}, \boldsymbol{\tau}_\ell)\right] \approx \arg\min_{\boldsymbol{\tau}_\ell^u \in \mathcal{T}_\ell^U} \sum_{u'=1}^{U} \hat{w}_{u'}\, d(\boldsymbol{\tau}_\ell^u, \boldsymbol{\tau}_\ell^{u'}),$$

where $d(A, B) = \max\{D_w(A, B), D_w(B, A)\}$ and
$D_w(A, B) = \frac{1}{|A|} \sum_{a \in A} \min\{|a - B|, h_d\}$. Here h is tolerance.

3. Interval Estimate: $[\alpha/2, 1 - \alpha/2]$ quantiles of $\mathcal{V}_k$. For each $\hat{\tau}_k \in \hat{\boldsymbol{\tau}}_{\mathrm{med}}$, define the valid match set $\mathcal{V}_k$ across B samples:

$$\mathcal{V}_k = \left\{ \arg\min_{t \in \boldsymbol{\tau}_\ell^{(b)}} |t - \hat{\tau}_k| \quad \text{s.t.} \quad \min |t - \hat{\tau}_k| \le h_d \right\}_{b=1}^{B}$$

- **Markov random field prior**

1. Match Score: Let $\mathcal{N}(\ell)$ be the set of spatial neighbors for line ℓ. We define a Match Score $m_\ell(u) \in [0, 1]$ for a candidate change point $u \in \boldsymbol{\tau}_\ell$:

$$m_\ell(u) = \mathbb{I}\left(\exists \ell' \in \mathcal{N}(\ell), \exists v \in \boldsymbol{\tau}_{\ell'} \text{ s.t. } |u - v| \le h\right)$$

where h is a bandwidth hyperparameter.

2. MRF Log-Potential: The spatial prior rewards supported points and penalizes isolated ones:

$$\log \pi_{\mathrm{MRF}}(\boldsymbol{\tau}_\ell \mid \{\boldsymbol{\tau}_{\ell'}\}_{\ell' \in \mathcal{N}(\ell)}) \propto \sum_{u \in \boldsymbol{\tau}_\ell} \Big(\underbrace{a_r \cdot m_\ell(u)}_{\text{Reward Alignment}} \underbrace{- a_p \cdot [1 - m_\ell(u)]}_{\text{Penalize Isolation}} \Big).$$

- **Red-black block Gibbs sampler**

1. Partitioning: Split lines into disjoint sets, e.g., **Red ($\mathcal{R}$, odd lines)** and **Black ($\mathcal{B}$, even lines)**:

$$\mathcal{R} = \{1, 3, 5, \dots\}, \quad \mathcal{B} = \{2, 4, 6, \dots\}$$

Neighbors are simply the lines above and below ($\mathcal{N}(\ell) = \{\ell - 1, \ell + 1\}$).

2. Alternating Updates (Parallelizable):

- Update Red: Sample all red lines in parallel conditional on black lines.
- Update Black: Sample all black lines in parallel conditional on red lines.

3. Modified RJMCMC Acceptance: The MRF introduces a spatial factor alongside the standard base prior:

$$\alpha = \min\left\{1, \underbrace{\frac{p(\mathbf{Y}_\ell \mid \boldsymbol{\tau}_\ell')}{p(\mathbf{Y}_\ell \mid \boldsymbol{\tau}_\ell)}}_{\text{Likelihood}} \times \underbrace{\frac{p(\boldsymbol{\tau}_\ell' \mid K')p(K')}{p(\boldsymbol{\tau}_\ell \mid K)p(K)}}_{\text{Base Prior (Sparsity)}} \times \underbrace{\frac{q(\boldsymbol{\tau} \mid \boldsymbol{\tau}')}{q(\boldsymbol{\tau}' \mid \boldsymbol{\tau})}}_{\text{Proposal}} \times \underbrace{\exp(\Delta \log \pi_{\mathrm{MRF}})}_{\text{Spatial Coupling}}\right\}$$

- **Vector field reconstruction**

- **Gradient Synthesis:** β is a directional derivative, i.e., $\beta_{dir} = \nabla E \cdot \boldsymbol{u}_{dir}$. Gradient reconstruction via vector summation

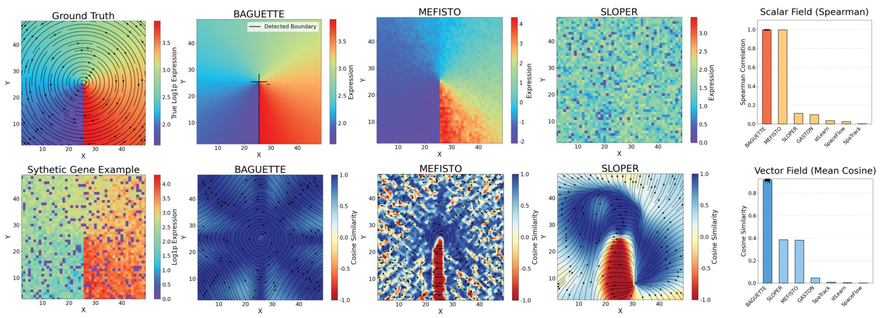
$$\mathbf{V} = \sum \beta_{dir} \mathbf{u}_{dir} = 1.5 \nabla E.$$

- **Spatial Smoothing (Component-wise LOESS):** Locally Estimated Scatterplot Smoothing (LOESS) was applied to the vector components (dx, dy) independently

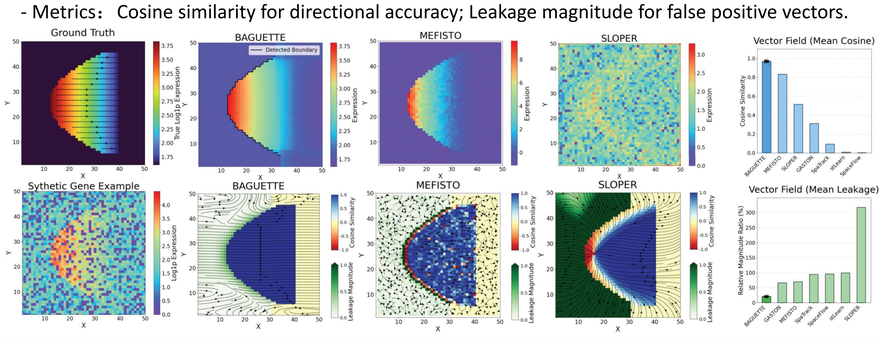
$$\mathbf{V}_{\mathrm{smooth}}(x, y) = (\mathrm{LOESS}(dx \sim x + y),\ \mathrm{LOESS}(dy \sim x + y))$$

3. SIMULATION

- **Simulation 1 (Vortex Field with Branch Cut):** Spurious Boundaries vs. Smoothness Assumption
  - Metrics：Spearman correlation for expression estimation; Cosine similarity for gradient estimation.

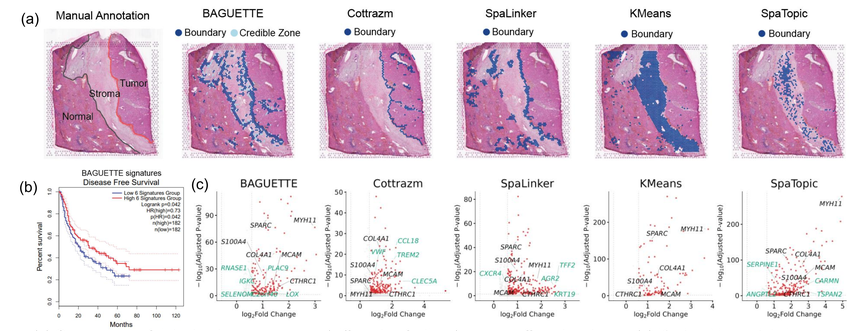


- **Simulation 2 (Open Hard Boundary):** Information Bleed-through across discontinuities
  - Metrics：Cosine similarity for directional accuracy; Leakage magnitude for false positive vectors.



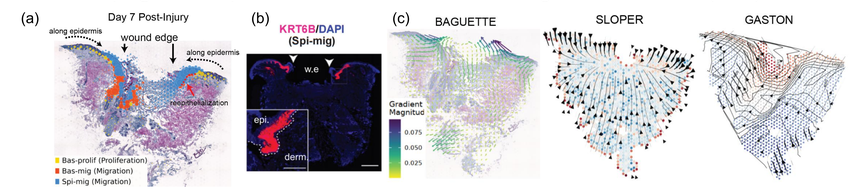
4. CASE STUDY

- **Case 1 (Tumor Boundary)**: 10x Visium hepatocellular carcinoma (HCC) from Wu et al. (2021).



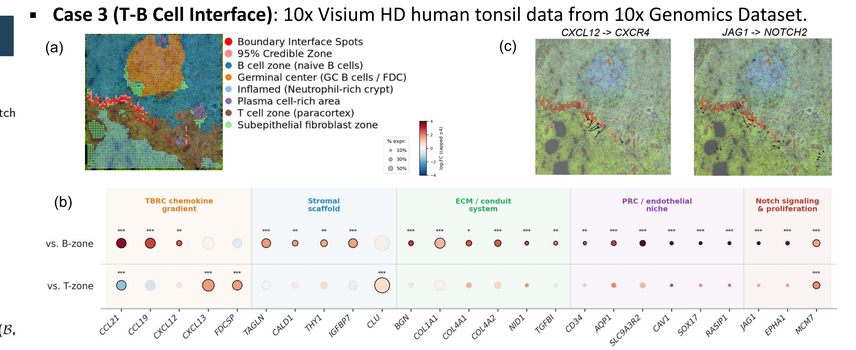
(a) Comparison of detected tumor boundary (different definitions) across different methods. (b) Kaplan-Meier plot shows prognostic value of BAGUETTE-identified spatial signatures in HCC. (c) Volcano plots for various methods. Consensus and unique markers are indicated in black and cyan, respectively.

- **Case 2 (Regeneration)**: 10x Visium human skin wound healing data from Liu et al. (2025).



(a) Illustration of the active re-epithelialization process. The spatial distribution of keratinocyte states reveals cells migrating along the epidermis and invading the wound bed at the wound edge to initiate tissue repair. (b) The Immunofluorescence validates a biological boundary between the epidermis and dermis. (c) Comparative analysis of vector fields. BAGUETTE highlights a physically consistent characterization of the directional cellular movements during re-epithelialization.

- **Case 3 (T-B Cell Interface)**: 10x Visium HD human tonsil data from 10x Genomics Dataset.



(a) Domain annotation adapted from STAHD. (b) Differential expression analysis indicates that the identified region represents a T-B interface-associated, TBRC(T–B border reticular cells)-like stromal niche, aligning with the fibroblastic niche framework proposed by De Martin et al. (2023). (c) The boundary-enriched *CXCL12–CXCR4* signaling functionally validates this region as a T–B interface-associated stromal niche, while the concurrent enrichment of *JAG1–NOTCH2* indicates specialized Notch-mediated niche specification and potential spatial remodeling.

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